

ML Lab

Visual standards audit and primitive strategy

A comparative review against Distill, Seeing Theory, R2D3 and other interactive-explanation benchmarks.

SCOPE

Homepage + 15 live exhibits

PREPARED

25 June 2026

PRODUCT

ml-lab-xi.vercel.app



A strong learning product that now needs a stronger visual identity.

ML Lab is already coherent, credible and pedagogically distinctive. Its four-act model - **See it, Run it, Break it, Explain it** - is a real advantage, not decorative navigation.

The project is not yet a consistently exceptional visual-essay product. Its shared page template is often more memorable than the individual exhibits, and too many ideas resolve into a cream panel containing plots, labels, legends and a metric strip.

CURRENT POSITION

Strong visual register 3: polished and instructional, but generally one level below the most authored work from Distill, Ciechanowski and R2D3.

BEST EDGE

Four-act pedagogy + genuine failure experiments

MAIN GAP

Template consistency has crossed into visual sameness

DESIGN PRIORITY

Fewer panels, stronger object continuity, visible causality

Every exhibit should contain one screenshot that could not plausibly belong to any other exhibit.

REPORT MAP

- 03 Competitive position
- 05 System strengths and gaps
- 06 Page-by-page audit
- 14 Primitive library
- 16 Priority roadmap

01 See it Form the mental model	02 Run it Inspect the implementation	03 Break it Learn the operating envelope	04 Explain it Prove transfer
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ML Lab wins on repeatable pedagogy. Competitors win on authored visual worlds.

The relevant benchmark is not visual polish alone. It is the degree to which form, interaction and explanation become inseparable.

INTEGRATED ARGUMENT

Distill

BENCHMARK STRENGTH

Prose, notation, annotation and interaction behave as one argument. The best pieces use linked representations and details-on-demand rather than placing an application beside an article.

IMPLICATION FOR ML LAB

ML Lab approaches Distill's semantic discipline, but trails in bespoke composition and section-by-section choreography. ML Lab is stronger as a repeatable learning product.

IMMEDIATE MANIPULATION

Seeing Theory

BENCHMARK STRENGTH

Readers touch the central object immediately: drag a point and see the OLS line, parameters, table and error respond together.

IMPLICATION FOR ML LAB

ML Lab looks more contemporary and has far stronger failure and transfer structure. Seeing Theory can still feel more immediate because the plot gets more space and carries less product chrome.

PERSISTENT VISUAL WORLD

R2D3

BENCHMARK STRENGTH

One visual world persists while data changes representation, flows through a tree and accumulates an argument.

IMPLICATION FOR ML LAB

ML Lab has stronger modular pedagogy, but weaker continuity. Its acts can feel like adjacent rooms rather than stages in one transformation.

REUSABLE PRODUCT SYSTEM

BESPOKE VISUAL ESSAY

ML Lab sits toward the system end. The strategic opportunity is to retain that reliability while giving every exhibit one unmistakable visual identity.

The largest gaps are objecthood, whole-system causality and emotional memory.

TACTILE OBJECTHOOD

Ciechanowski

BENCHMARK STRENGTH

The subject becomes an object: drag it, rotate it, cut it open, then progressively decompose and reconstruct it.

IMPLICATION FOR ML LAB

This is the largest gap. ML Lab remains predominantly flat and chart-like even when the concept naturally supports terrain, flow, tension or spatial transformation.

WHOLE-SYSTEM VISIBILITY

TensorFlow Playground

BENCHMARK STRENGTH

Inputs, hidden units, connections, decision field, loss and training are visible together on one live bench.

IMPLICATION FOR ML LAB

ML Lab is substantially better at onboarding and explanation, but less effective at showing the complete causal system at once.

PLAY AND MEMORY

Nicky Case / Explorables

BENCHMARK STRENGTH

Play, personality, feedback and game progression create emotional memory as well as conceptual understanding.

IMPLICATION FOR ML LAB

ML Lab is more restrained and academically rigorous, but its interactions rarely feel surprising, rewarding or playful.

STORY-SPECIFIC IDENTITY

The Pudding

BENCHMARK STRENGTH

Each story earns its own editorial voice, metaphor and pacing rather than conforming to a publication-wide page template.

IMPLICATION FOR ML LAB

ML Lab has greater system coherence; The Pudding has greater story identity.

BORROW FROM DISTILL	BORROW FROM R2D3	BORROW FROM CIECHANOWSKI	KEEP FROM ML LAB
Annotation at the point of meaning	One protagonist that changes over time	Tactile and spatial metaphors	Failure diagnosis and transfer

Preserve the semantics. Loosen the composition.

WHAT IS ALREADY WORKING

- 01

Semantic visual grammar

Prediction blue, truth gold, error red and parameter purple mean the same thing throughout, including in prose.
- 02

Break it as a differentiator

Learners cross the operating boundary, observe a symptom, diagnose the cause and repair the system.
- 03

Mechanism-bearing exhibits

Loss Functions, Feature Scaling, Bias-Variance, Data Leakage and The Dataset already put cause and consequence in the picture.
- 04

Recognisable front door

The jewel cabinet and graph relationships create more intentional navigation than a conventional catalogue.

WHAT IS HOLDING IT BACK

- 01

Template tax

Hero, placard, act rail, prose column, panel, metric strip and field guide recur in nearly identical arrangements.
- 02

Diptych saturation

Before/after is useful, but too often replaces one continuous, reversible transformation.
- 03

Flat objecthood

Concepts are diagrammed rather than embodied as terrain, flow, tension, folding or spatial change.
- 04

Persistent explanation chrome

Legends, metric strips and side cards remain visible after their role has been learned.
- 05

Weak act continuity

The protagonist rarely travels visibly from See to Run, becomes damaged in Break and is reconstructed in Explain.

SIX DESIGN PRINCIPLES

1 One unmistakable frame per exhibit	2 One protagonist across all four acts	3 Put the mechanism in the picture
4 Reveal explanation at the point of need	5 Use motion to expose causality	6 Keep semantics consistent; vary composition

Front door / Groundwork

01-02 OF 16

01 FRONT DOOR

Homepage

CURRENT VISUAL READ

The jewel cabinet is distinctive and more intentional than a conventional course catalogue. Its abstract dark gems, however, do not preview the warm, diagrammatic and interactive experiences behind them.

HIGHEST-VALUE NEXT MOVE

Place a live micro-specimen inside every jewel, then visually morph the selected jewel into the exhibit hero. The homepage should demonstrate the product, not only describe it.

[MicroSpecimen](#)[ActHandoff](#)

02 GROUNDWORK

What is ML?

CURRENT VISUAL READ

The 70% hand rule versus 97% learned rule carries the thesis well. Dot weight, label collisions and lower-left emptiness reduce the authority of the opening frame.

HIGHEST-VALUE NEXT MOVE

Use one reversible counterfactual morph from hand rule to learned rule. Keep mistakes as persistent ghosts so the learner can see exactly what changed and what was repaired.

[CounterfactualReplay](#)[CausalTrace](#)

Groundwork / First models

03-04 OF 16

03 GROUNDWORK

The Dataset

CURRENT VISUAL READ

One mistyped row visibly drags the fitted line. This is among the strongest causal demonstrations in the collection, with particularly disciplined colour and an honest cause-to-effect story.

HIGHEST-VALUE NEXT MOVE

Add a permanent row-to-point tether and a provenance lens that reveals where each displayed value originated. Make the bad datum feel traceable, not merely highlighted.

[ProvenancePipe](#)[RepresentationPortal](#)

04 FIRST MODELS

Regression Task

CURRENT VISUAL READ

Residual-as-distance is correctly shown, but the hero remains a relatively generic scatter-and-line composition. The visual says "regression" more clearly than it says "the task and its score."

HIGHEST-VALUE NEXT MOVE

Introduce an error ruler and let individual residuals physically accumulate into the total metric. The score should visibly be made from the misses.

[ContributionStack](#)[ProbeLens](#)

First models

05-06 OF 16

05 FIRST MODELS

Linear Regression

CURRENT VISUAL READ

A solid full-width specimen with residual ticks and good in-canvas annotation. The result is more visible than the minimisation process that makes the line special.

HIGHEST-VALUE NEXT MOVE

Turn residuals into areas that visibly combine into MSE, then let learners scrub candidate lines through the same persistent canvas. Make "best" an observable search, not a label.

[ContributionStack](#)[PinAndCompare](#)

06 FIRST MODELS

Classification Task

CURRENT VISUAL READ

Threshold, outcomes, confusion matrix and metrics are all present, but the hero is busy and dashboard-like. The learner must mentally connect several adjacent representations.

HIGHEST-VALUE NEXT MOVE

Use a decision conveyor: observations cross the threshold and physically land in TP, FP, FN and TN bins while precision and recall update.

[DecisionConveyor](#)[CausalTrace](#)

First models / How models learn

07-08 OF 16

07 FIRST MODELS

Logistic Regression

CURRENT VISUAL READ

The probability field and trained boundary are effective. The untrained panel is comparatively empty, and the relationship between the 0.5 probability crossing and the 2D boundary is not yet the protagonist.

HIGHEST-VALUE NEXT MOVE

Link the 2D field to a movable 1D sigmoid cross-section. The 0.5 crossing should be physically attached to the boundary and move with it.

[RepresentationPortal](#)[ProbeLens](#)

08 HOW MODELS LEARN

Loss Functions

CURRENT VISUAL READ

One of the strongest exhibits. Penalty area is the picture: outliers visibly dominate squared error and the mechanism does not depend on explanatory prose.

HIGHEST-VALUE NEXT MOVE

Show the same miss physically deforming under L1, L2 and Huber instead of relying on separate lines and legends. Preserve one object while the judge changes.

[CounterfactualReplay](#)[ContributionStack](#)

How models learn

09-10 OF 16

09 HOW MODELS LEARN

The Gradient

CURRENT VISUAL READ

The contour landscape and perpendicular arrow are immediately legible, but the experience remains visually flat. The learner sees the gradient without fully feeling local slope.

HIGHEST-VALUE NEXT MOVE

Add a draggable probe lens that reveals the local tangent plane, directional slopes and gradient components at the point of inspection.

[ProbeLens](#)[RepresentationPortal](#)

10 HOW MODELS LEARN

Gradient Descent

CURRENT VISUAL READ

A strong contour poster and a controlled descent path make this one of the better hero compositions. The full path is clear; the anatomy of one update is still compressed.

HIGHEST-VALUE NEXT MOVE

Add a step microscope that freezes one update and decomposes gradient, parameter delta, momentum and loss change before resuming the path.

[StepMicroscope](#)[CausalTrace](#)

How models learn / Keeping it honest

11-12 OF 16

11 HOW MODELS LEARN

Feature Scaling

CURRENT VISUAL READ

Probably the strongest visual explanation in the collection: surface geometry and optimizer behaviour appear together. The before/after diptych is effective but still splits one causal transformation into two panels.

HIGHEST-VALUE NEXT MOVE

Replace the diptych with an axis-deformation morph in which the same loss surface visibly stretches from a narrow valley into a bowl while the optimizer path straightens.

[CounterfactualReplay](#)[ActHandoff](#)

12 KEEPING IT HONEST

Train-Test Generalization

CURRENT VISUAL READ

The lottery histogram and train/CV pins are clear and well labelled. It reads more as a finished statistical chart than as an experiment whose distribution the learner helped create.

HIGHEST-VALUE NEXT MOVE

Deal observations into train and test sets like cards, then let repeated split results accumulate into the histogram. Make uncertainty visibly emerge from repeated action.

[VarianceSwarm](#)[CausalTrace](#)

Keeping it honest

13-14 OF 16

13 KEEPING IT HONEST

Bias-Variance

CURRENT VISUAL READ

The degree triptych and error U expose the mechanism effectively. Three snapshots communicate the states, but they underplay instability across possible training samples.

HIGHEST-VALUE NEXT MOVE

Use one complexity scrubber and a variance swarm of models trained on resamples. The learner should see the family of possible curves spread, not only one representative curve.

[VarianceSwarm](#)[PinAndCompare](#)

14 KEEPING IT HONEST

Overfitting and Regularization

CURRENT VISUAL READ

Lambda, the error curve and coefficient magnitudes work well, but the exhibit remains visually close to Bias-Variance. The intervention needs a more distinctive material metaphor.

HIGHEST-VALUE NEXT MOVE

Add a shrinkage trace: coefficients contract, the curve is pulled smooth and test error responds in one chained movement. Regularization should feel like visible tension.

[CausalTrace](#)[RepresentationPortal](#)

Keeping it honest / Going deeper

15-16 OF 16

15 KEEPING IT HONEST

Data Leakage

CURRENT VISUAL READ

The leaked versus honest R-squared contrast is conceptually strong, but the twin scatterplots are too similar. The forbidden information flow is more important than either final plot.

HIGHEST-VALUE NEXT MOVE

Make a provenance pipe the protagonist: forbidden information visibly crosses the train/test wall and contaminates the score. Repairing the boundary should visibly clean the pipeline.

[ProvenancePipe](#)[CounterfactualReplay](#)

16 GOING DEEPER

Neural Network Fundamentals

CURRENT VISUAL READ

XOR failure versus success is clear, but the hidden layer itself is not yet the visual protagonist. The learner sees a bent boundary without seeing how hidden units made it possible.

HIGHEST-VALUE NEXT MOVE

Show hidden units as individual half-space transformations that progressively fold feature space into a separable arrangement. Let learners add, mute and inspect each fold.

[RepresentationPortal](#)[StepMicroscope](#)

Compare, trace and inspect

● Compare ● Inspect ● Carry ● Identity

These should be state-driven visual-argument components, not additional cards. Every primitive should inherit the existing semantic colours, work under reduced motion and remain legible as a static frame.

COMPARE AND INTERVENE

01

CounterfactualReplay

Two synchronized worlds share one seed and state; exactly one intervention differs. Playback is reversible.

FIRST APPLICATIONS Scaling, regularization, leakage

TRACE AND INSPECT

02

CausalTrace

A brief pulse traces from the manipulated cause through intermediate representations to the final metric.

FIRST APPLICATIONS Dataset, regularization, classification

TRACE AND INSPECT

03

RepresentationPortal

Hovering an entity in one representation highlights the corresponding entity everywhere else.

FIRST APPLICATIONS Regression, loss, gradients, networks

TRACE AND INSPECT

04

StepMicroscope

Freeze one learning step and expose the before-state, vector, update and after-state.

FIRST APPLICATIONS Gradient descent, neural networks

TRACE AND INSPECT

05

ProbeLens

Reveal local detail on demand without leaving permanent labels and legends on the canvas.

FIRST APPLICATIONS Gradient, logistic regression

COMPARE AND INTERVENE

06

PinAndCompare

Pin the current state as a ghost, then compare a new state directly or view only the difference.

FIRST APPLICATIONS Thresholds, lambda, degree

COMPARE AND INTERVENE

07

ContributionStack

Individual example contributions visibly aggregate into total loss or score.

FIRST APPLICATIONS Regression task, linear regression, loss

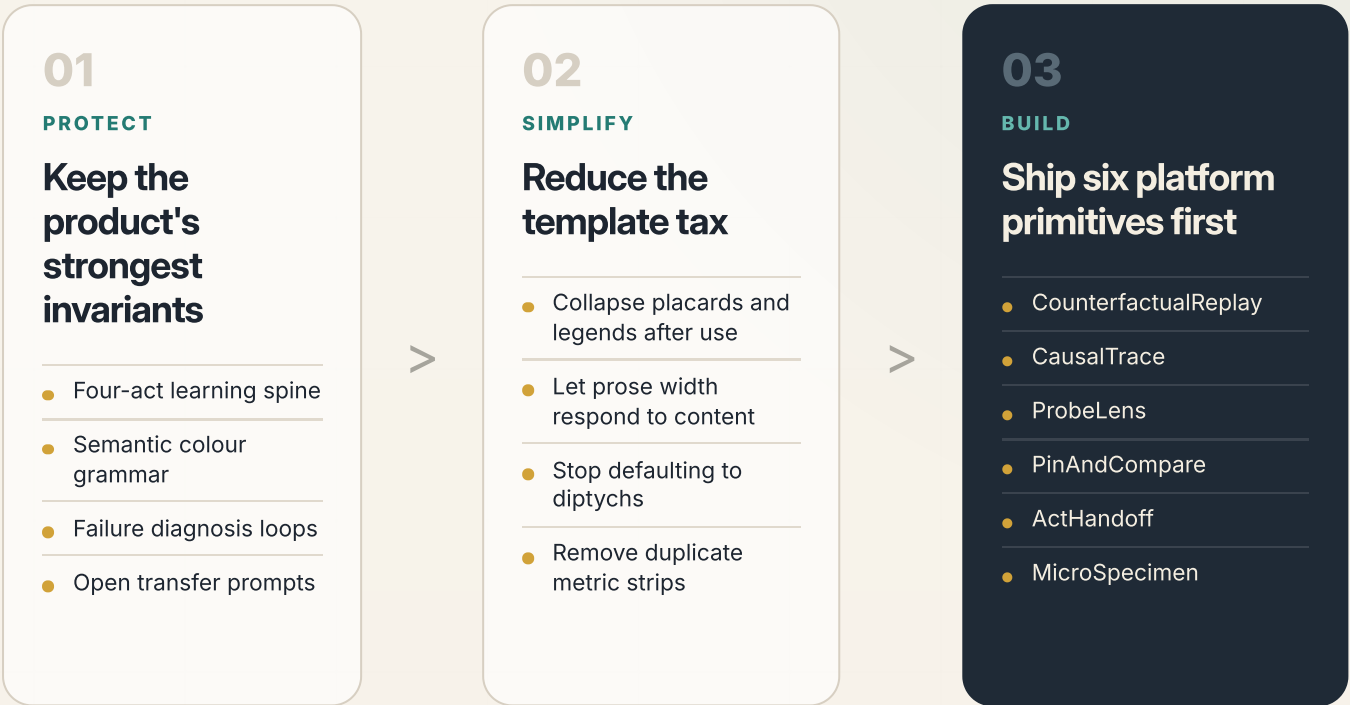
Connect, carry and create identity

● Compare ● Inspect ● Carry ● Identity

These should be state-driven visual-argument components, not additional cards. Every primitive should inherit the existing semantic colours, work under reduced motion and remain legible as a static frame.

CONNECT AND CARRY ProvenancePipe Represent collection, transformation, split and evaluation as constrained data flow with visible boundary violations. FIRST APPLICATIONS Dataset, leakage, train-test	08	COMPARE AND INTERVENE DecisionConveyor Observations move through a rule and settle into outcome bins while metrics update. FIRST APPLICATIONS Classification, logistic regression	09
CONNECT AND CARRY VarianceSwarm Repeated samples or trained models accumulate as a swarm, envelope or distribution. FIRST APPLICATIONS Bias-variance, train-test	10	CONNECT AND CARRY ExplorableSentence A word, value or phrase in the prose is itself the control and highlights the corresponding mark. FIRST APPLICATIONS Every See act	11
CONNECT AND CARRY ActHandoff The final state and protagonist from one act visibly become the starting point of the next. FIRST APPLICATIONS Platform-wide	12	IDENTITY AND NAVIGATION MicroSpecimen A restrained miniature of the exhibit's defining transformation, usable without full controls. FIRST APPLICATIONS Homepage and journeys	13
IDENTITY AND NAVIGATION ExhibitSkin One concept-specific material, geometric motif and signature transition layered over the global semantics. FIRST APPLICATIONS Platform-wide	14		

Remove sameness first. Then add a small number of high-leverage primitives.



RECOMMENDED PILOT SEQUENCE

1 Data Leakage Provenance pipe	2 Neural Network Fundamentals Feature-space folding	3 Classification Task Decision conveyor
4 Gradient Descent Step microscope	5 Feature Scaling Continuous axis morph	6 Homepage Live micro-specimens

PROPOSED ACCEPTANCE TESTS

Shuffled screenshot Can a reviewer identify the exhibit without seeing its title?	Causal path Can the learner point from input to mechanism to consequence?	Act continuity Does the same protagonist visibly survive all four acts?	Chrome discipline Does every persistent label still earn its space?
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The direction is not more visualizations. It is fewer panels, stronger object continuity and visible causality.

Audit scope and source set

SCOPE

Homepage + every live exhibit

Sixteen surfaces were assessed: the homepage and 15 exhibit routes across the full Foundations journey.

SURFACES

Hero, See, Run, Break, Explain

The audit considered opening composition, guided story states, open experiments, failure loops and transfer surfaces.

COMPARISON

Named interactive-explanation benchmarks

Competitors were compared for visual argument, object continuity, direct manipulation, whole-system visibility and editorial identity.

QUALIFICATION

Comparative design review

This is an expert visual and interaction audit, not a formal user study or accessibility certification.

PRIMARY REFERENCES

ML Lab

Public product

ml-lab-xi.vercel.app

ML Lab repository

Exhibit registry, page system, visual tokens and review scorecards

github.com/cbrown564-alt/ml-lab

Distill

Communicating with Interactive Articles

distill.pub/2020/communicating-with-interactive-articles

Seeing Theory

Regression Analysis

seeing-theory.brown.edu/regression-analysis/index.html

R2D3

A Visual Introduction to Machine Learning

r2d3.us/visual-intro-to-machine-learning-part-1

Bartosz Ciechanowski

Mechanical Watch

ciechanow.ski/mechanical-watch

TensorFlow Playground

Interactive neural-network playground

playground.tensorflow.org

Nicky Case

The Evolution of Trust

ncase.me/trust

The Pudding

How to Make Dope Shit, Part 3

pudding.cool/process/how-to-make-dope-shit-part-3

BOTTOM LINE

ML Lab already has the hard part: a coherent educational model and honest interactive mechanics. The next step is to make each concept feel like its own world without losing the system beneath it.